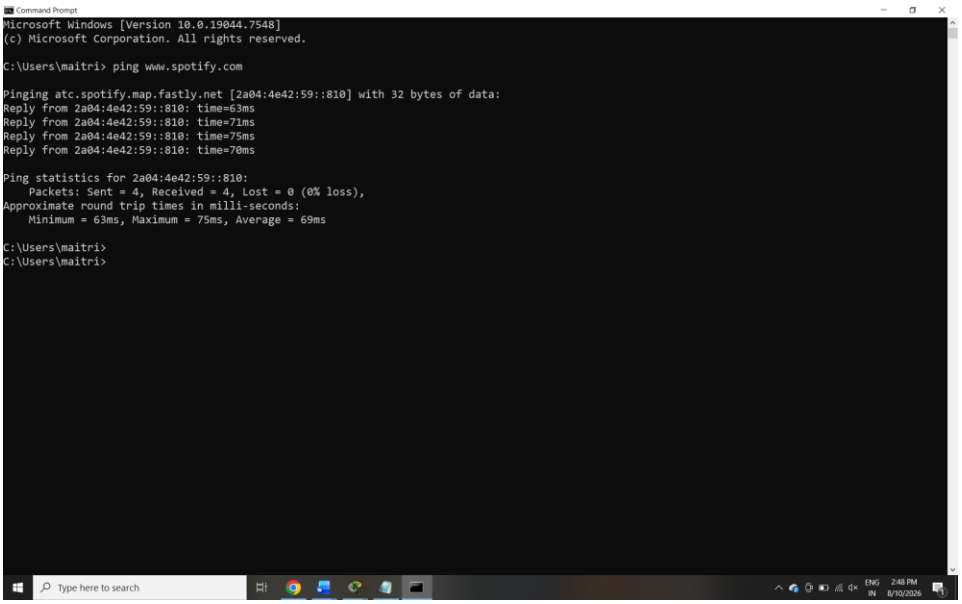
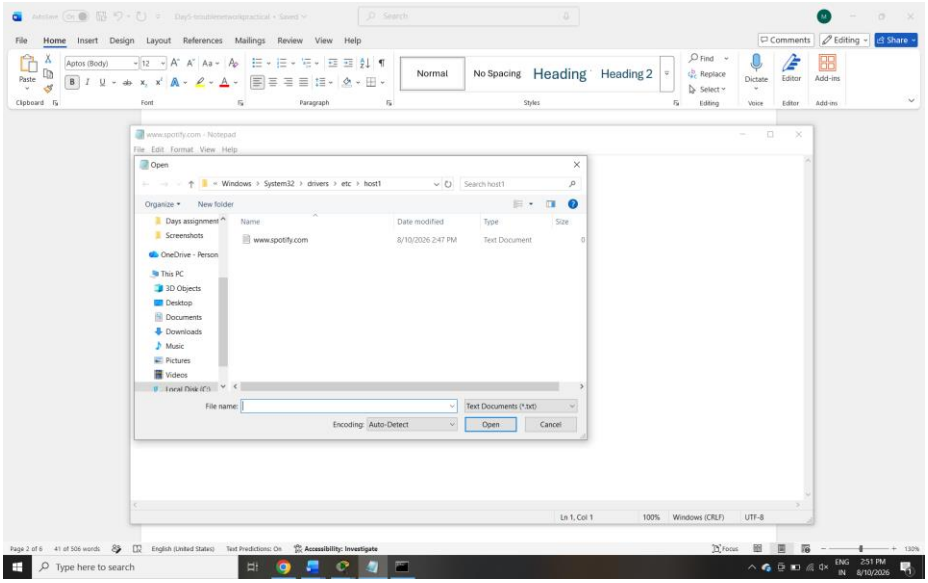
1.Simulate a DNS resolution failure on your local machine by editing the hosts file to point www.spotify.com to a wrong IP address, then try to ping the domain and document the error message you receive.





2.Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.

1. Add 1 Router, 1 Switch, 1 Server, and 1 PC in Cisco Packet Tracer and connect them.
2. Set Server IP: `192.168.10.2` and Subnet Mask: `255.255.255.0`.
3. Go to Server → Services → DHCP → ON.
4. Create a DHCP pool:
    - Start IP: `192.168.10.10`
    - Subnet Mask: `255.255.255.0`
    - Default Gateway: `192.168.20.1` (wrong)
5. Set the PC to DHCP and run `ipconfig`.

Example:

```
IP Address: 192.168.10.10
Default Gateway: 192.168.20.1
```

Problem: The gateway is in a different network.

Fix: Change the DHCP gateway to 192.168.10.1, renew DHCP on the PC, and test with:

```
ping 192.168.10.1
```

The ping should work after the gateway is corrected.

3.Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the nslookup or dig tool to diagnose the DNS problem and write down the troubleshooting steps and solution.

```
Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : 30-F6-EF-CA-88-F3
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : 32-F6-EF-CA-88-F2
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Wi-Fi 6 AX201 160MHz
   Physical Address. . . . . . . . . : 30-F6-EF-CA-88-F2
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2401:4900:88b6:2ae8:cd10:61f3:5211:365f(Preferred)
   Temporary IPv6 Address. . . . . . : 2401:4900:88b6:2ae8:898b:4179:74d8:dcd5(Preferred)
   Link-local IPv6 Address . . . . . : fe80::ac07:9e95:6e2e:6aca%18(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.1.4(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 08 August 2026 13:08:02
   Lease Expires . . . . . . . . . . : 09 August 2026 13:08:02
   Default Gateway . . . . . . . . . : fe80::4663:c2ff:fe34:3cc0%18
                                       192.168.1.1
   DHCP Server . . . . . . . . . . . : 192.168.1.1
   DHCPv6 IAID . . . . . . . . . . . : 137426671
```

```
C:\Users\jurmi>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=52ms TTL=117
Reply from 8.8.8.8: bytes=32 time=48ms TTL=117
Reply from 8.8.8.8: bytes=32 time=46ms TTL=117
Reply from 8.8.8.8: bytes=32 time=46ms TTL=117

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 46ms, Maximum = 52ms, Average = 48ms
```

```
C:\Users\jurmi>nslookup www.google.com
Server:  UnKnown
Address:  fe80::4663:c2ff:fe34:3cc0

Non-authoritative answer:
Name:    www.google.com
Addresses:  2001:4860:482a:7700::
          2001:4860:482c:7700::
          2001:4860:4826:7700::
          2001:4860:4829:7700::
          2001:4860:482b:7700::
          2001:4860:482d:7700::
          2001:4860:4828:7700::
          2001:4860:4827:7700::
          142.251.156.119
          142.251.154.119
          142.251.155.119
          142.251.151.119
          142.251.150.119
          142.251.157.119
          142.251.153.119
          142.251.152.119
```

**4.A device in your simulated network is not receiving an IP address from the DHCP server and shows an APIPA address (169.x.x.x). List the possible causes for this and document the step-by-step process you would use to resolve it.

Hint: Check for network connectivity, DHCP scope exhaustion, and server availability.**

Possible Causes

- DHCP server is turned off.
- Network cable is disconnected.
- Wrong IP/DHCP configuration.
- DHCP address pool is full.
- Device is connected to the wrong network/VLAN.

Steps to Resolve

1. Check the cable connection.
2. Make sure the DHCP server is ON.
3. Check the DHCP pool and ensure free IP addresses are available.
4. Set the device to Obtain IP automatically (DHCP).
5. Renew the IP address (`ipconfig /renew` or restart the device).

Result: The APIPA address (169.x.x.x) should change to a valid IP address from the DHCP server (for example, 192.168.1.100).

5.Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.

Common DNS and DHCP Issues

Protocol / Issue	Symptoms	Possible Causes	Recommended Solution
DNS Issue 1: Website name not resolving	Websites do not open using their names (e.g., google.com), but may work with an IP address.	Incorrect or unreachable DNS server.	Configure a valid DNS server (such as 8.8.8.8) and verify network connectivity.
DNS Issue 2: Slow website loading	Web pages take a long time to start loading.	Slow DNS server response or overloaded DNS server.	Use a faster DNS server and clear the DNS cache.
DNS Issue 3: Wrong website opened	Typing the correct domain opens a different or incorrect website.	Incorrect DNS record, cached entry, or DNS poisoning.	Flush the DNS cache and correct the DNS records.
DHCP Issue 1: No IP address assigned	Device shows “No Internet”, “Limited connectivity”, or an APIPA address (169.254.x.x).	DHCP server is down or unreachable.	Restart the DHCP server/router and renew the IP address.
DHCP Issue 2: IP address conflict	Network warning about a duplicate IP address; intermittent connectivity.	Two devices are using the same IP address.	Release and renew the IP address or assign a different static IP.
DHCP Issue 3: Wrong network configuration	Device gets an IP address but cannot access the internet.	Incorrect DHCP settings (gateway, subnet mask, or DNS).	Correct the DHCP scope/options and renew the client configuration.